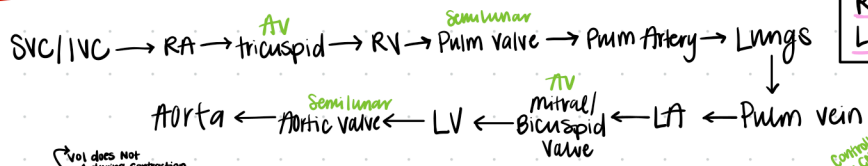
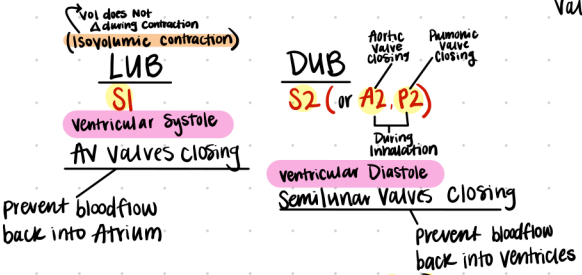
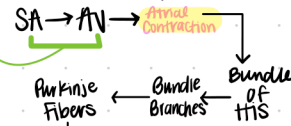


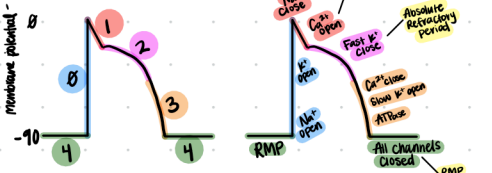
RAT LAB



Electrical impulse:



Cardiac RMP = -90mV



- Phase 4: RMP (more Na⁺ outside, more K⁺ inside)
- Phase 0: Depolarization
- Phase 1: Wait
- Phase 2: plateau
- Phase 3: Repolarization

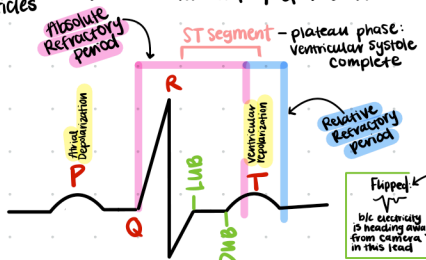
ARP

RRP

ECG/EKG -

recording electrical activity that then results in contraction/relaxation

Electrical impulses result from waves of the depol/repol of AP



Bipolar leads:



Electrode is camera. As electricity heads toward camera, electrical potential ↑

Unipolar leads:

- AVR - R arm
 - AVL - L arm
 - AVF - L leg (foot)
- Chest/precordial - V₁ - V₆

Lymphocytes - Derived from stem cells in bone marrow

Primary Lymphoid Organs:
Bone Marrow
Thymus
↳ Site of new Lymphocytes

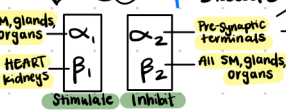
Secondary Lymphoid Organs:
Lymph nodes
Spleen - filters blood for pathogens
Tonsils
Peyer's patches

Blood reservoir:
venous side
most of the blood is here @ any given time

Pressure reservoir:
Arterial side
BO is the floor, Arteries provide some amt of pressure on their own

Cholinergic receptors: Bind ACh
↳ Muscarinic - on ♥ & vessels
↳ Nicotinic - on skeletal muscle

Adrenergic receptors: Bind NE
A, B, C, D α = Constrict
β = Dilate



Also in blood vessels: tells them to relax

Negative Feedback! So it inhibits the post-gang neurons → prevents over exaggeration

has to push ↑ higher pressure to get blood through = Raising BP
equal to vasoconstriction

LOW BP:
↑ SNS activity
↓ PSNS activity
HR ↑, TPR ↑

HIGH BP:
↓ SNS, ↑ PSNS
HR ↓, TPR ↓
equal to vasodilation

Cerebral circulatory response
is opposite SNS response elsewhere:
BPT → vasoconstriction
BP ↓ → vasodilation

pulse pressure =
Systolic - diastolic

Stroke volume =
amt of blood expelled from ♥ per systole (beat)
Normal = 30-40

Mean Arterial Pressure (MAP) =
1/3 (Systolic) + 2/3 (diastolic)

Baroreceptor = Stretch receptor
Activated by ↑ BP, increases ARs to vasomotor & cardiac centers in the medulla

to get more O₂ in

Chemoreceptors = respond to chemical Δ
(Δ in O₂, CO₂, pH)
inverse relationship - PCO₂ = ↓ pH

Cerebral circulatory response:
CO₂ build-up (↓ pH) → vasodilation
CO₂ drop (↑ pH) → vasoconstriction



Cardiac Output =
Vol pumped each
min by each ventricle

$$CO = HR \times SV$$

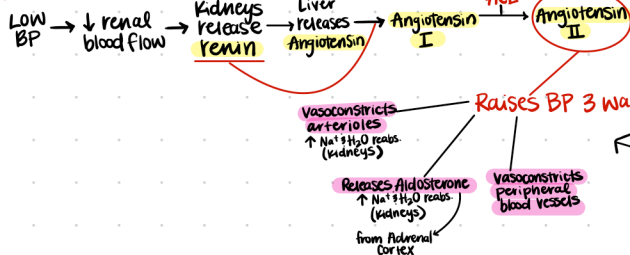
$$SV = (\text{Contractility} + \text{preload}) - \text{afterload}$$

Strength of
contraction
(inotropy) $\uparrow$
inotropy $\uparrow$
SV $\uparrow$

EDV $\uparrow$
SV $\uparrow$
Determined
by
venous return
End Diastolic Vol
EDV
Vol left in
ventricles
@ end of diastole

TPR
Total peripheral
resistance
TPR $\uparrow$
SV $\downarrow$
impedes ejection
of blood from
ventricle

RAAS $\rightarrow$



BP Regulation $\rightarrow$

OSmoreceptors detect $\uparrow$ plasma osmolarity (Na^+ concn)
 $\rightarrow$ can $\uparrow$ due to $\uparrow$ Na^+ intake OR $\downarrow$ water intake
ADH (Vasopressin) released from post. pituitary
Causes $\uparrow$ thirst
Stimulates H_2O reabsorption in kidneys
 $\rightarrow$ stimulates $\downarrow$ output
Brings up blood vol $\&$ BP

Stretch receptors in Atria of $\heartsuit$ detect

$\uparrow$ blood volume
Atrial Natriuretic Peptide (**ANP**)
released from atria
Causes kidneys to excrete
 H_2O and Na^+ (water follows)

Stimulates $\uparrow$ output
Brings down blood vol $\&$ BP
*** ANP makes you pee**

Aldosterone - secreted by Adrenal cortex
when blood vol $\&$ BP are reduced
stimulates reabsorption of Na^+ $\&$ water
in the kidneys
 $\rightarrow$ stimulates $\downarrow$ urine output
Brings up blood vol ($\uparrow$) $\&$ BP ($\uparrow$)

Regulated by RAAS

WBC

Hematopoiesis: All blood cells start in Bone Marrow
(multipotential hematopoietic stem cells)

Interleukin-3
Controls growth $\&$
reproduction
(growth inducer)

Leukocytes:

Granulocytes - Neutrophils
Eosinophils
Basophils

Agranulocytes - Lymphocytes
Monocytes $\rightarrow$ become Macrophages

Kupffer cells: Liver sinusoids macrophages
bacteria from ingested food goes into portal
blood - before entering circulation it passes
through Kupffer cells to filter antigens out

Inflammation

lines of defense:
1st Macrophages - begin phagocytizing
2nd Neutrophil invasion - Byproducts of inflammation
3rd Macrophage invasion Again - storage depleted
4th bone marrow $\uparrow$ production
of granulocytes $\&$ monocytes - committed cells
stimulated

Eosinophils - parasites stimulate their production
Also play role in inflammation
Associated $\&$ Allergic rxns

Basophils (in blood) - Similar to mast cells (outside the capillaries)
Both release histamine, bradykinin $\&$ Serotonin
(which makes inflammation worse)
 $\rightarrow$ Mast cells more likely to release these
They both play a big role in Allergic rxns
IgE is a class of antibodies that work $\&$ basophils
 $\&$ mast cells to cause allergic response

Releases a Leukotriene mixture:
Causes anaphylaxis $\&$ lysosomal enzymes
 $\rightarrow$ local vascular/tissue rxn (rashes)

- 1) Stay in bone marrow for backup
- 2) Committed Stem cells (not mature yet)
- 3) Differentiate

Float in amoeboid motion until
stimulated by chemotaxis
to move toward inflamed area
phagocytosis:
• Rough surface
• Protein coat absent
• Antibodies on surface
Do their job $\&$ then die

Antigen - toxin
Antibody - protection

Immunity:

Acquired (or adaptive) - develops $\&$ encountering
a specific bacteria/virus
Initiated by **Antigens**
 $\downarrow$
Antibody generators

2 types:

Humoral - B-cell
B lymphocyte Attack

Cell mediated - T-cell
T lymphocyte Destroy

Liver $\&$
bone marrow
Thymus

RBC

People $\in$ vitk
deficiency are
probably excessive
bleeders.

Clotting factors produced
in Liver by hepatocytes
EXCEPT Factor 8
(produced in Liver but Not
by hepatocytes)

Intinsic = Slow
 $\odot$ changed sub-
endothelial
substance/
Vascular injury

Extinsic = Explosive
TF

usually clotting
factors = inactive
but $\sim 1\%$ Factor 7
is the exception
(spontaneously
activated)

AT3 inhibits
Thrombin $\&$ Factors
7, 9, 11, 12
Extrinsic $\swarrow$ Intrinsic

Heparan sulfate
makes AT3 much
stronger!

Ventilation: Breathing

Gas exchange

- External Respiration - Ventilation/gas exchange in the lungs
- Internal Respiration - oxygen utilization & gas exchange b/w blood & tissues

occurs via **Simple diffusion**

- O_2 concentration higher in lungs than blood
- CO_2 concentration higher in blood than lungs

Respiratory system:

Conduction zone - transports air → warms, humidifies, filters & cleans air; mucus traps particles

Respiratory zone - Site of gas exchange
terminal bronchioles; **Alveoli**

Alveoli → Air sacs: Large SA for ↑ diffusion rate

Type 1 - SA for gas exchange

Type 2 - Secrete **Surfactant** Prevents alveoli collapse (reduces surface tension)

Elasticity: Lungs return to initial size & stretch (Recoil)

Lungs attached to thoracic wall = Always under elastic tension

→ Inc during inspiration
dec during expiration (by elastic recoil)

Breathing:

Inspiration - Diaphragm contracts/flattens
Accessory: Serratus, pec minor, SCM
External intercostals

Expiration - Quiet - Diaphragm relaxes
Forced - Internal intercostals
Abdominal muscles

	Inspiration (↓ intrapulm pressure)	Expiration (↑ intrapulm pressure)
Quiet breathing	-3	+3
Forced ventilation	-20	+30

*So as volume goes ↑, pressure goes ↓, v/v ↑

Breathing Regulation-

Voluntary control:

Cerebral cortex

Involuntary control:

Medulla & pons

Apneustic center - influences medulla
Inspiration
Pneumotaxic center - Expiration

Chemoreceptors:

monitor pH & CO_2 levels

Feedback controls automatic breathing

Increased CO_2 is the primary drive for breathing

→ Ventilation controlled to maintain constant CO_2 levels in the blood (pH 7.35-7.45)

Gas Exchange-

Dalton's Law:

Total gas pressure = sum of pressures of each gas in it

Partial pressure:

pressure of individual gas; % of gas x total pressure

PO_2 ; PCO_2

Determines Gas Exchange Direction

- Only measures O_2 dissolved in blood plasma - Does Not measure O_2 bound to hemoglobin in RBCs

- When functioning properly:

PO_2 of systemic arterial blood is only 5 mmHg less than

PO_2 of alveolar air → PO_2 Arteries = 100 mmHg

PO_2 Alveoli = 105 mmHg

Hypoventilation = ↑ CO_2 → ↓ pH

Hyperventilation = ↓ CO_2 → ↑ pH

hypocapnia

hypoxia

Ex. LOW O_2 = Carotid bodies become more sensitive to CO_2

O_2 level indirectly affects ventilation by affecting chemoreceptor sensitivity to CO_2

PFTs:

Spirometry - measures lung vol & capacities

↳ Dx restrictive/dysruptive lung disorders

FEV1 (Forced Expiratory Volume) - Measures % vital capacity (air vol) that can be forcibly exhaled in 1s & full inspiration

↳ Dx lung diseases

Pressure types:

Atmospheric - pressure outside body

Intrapulmonary - pressure inside lungs

Intrapleural - pressure within intrapleural space

(-) pressure to keep lungs from collapsing
↳ b/w parietal & visceral layers
↳ contains thin fluid = lubricant

Boyles Law: pressure inversely proportional to volume

When breathing:

INSPIRATION -

Intrapulmonary pressure LOWER than

Atmospheric pressure

EXPIRATION -

Intrapulmonary pressure greater than

Atmospheric pressure

→ Increase in lung vol then decreases intrapulm pressure
≡ Air goes in

→ Decrease in lung vol then increases intrapulm pressure
≡ Air goes out

Law of Laplace: pressure inversely proportional to alveoli radius

Large radius = lower pressure

Surfactant important here:

reduces surface tension; prevents collapse
Allows residual vol of air to stay in lungs

Lung Vol & Capacities:

Tidal Vol (TV) - Normal breathing volume (quiet breathing)

Inspiratory Reserve Vol (IRV) - Forceful inhaled above TV

Inspiratory Capacity - Amt that can be inspired & TV
TV + IRV

Expiratory Reserve Vol (ERV) - Forceful exhaled below TV

Residual Vol (RV) - Air left in lungs & max expiration

Functional Residual Capacity (FRC) - Air left in lungs & Normal expiration
ERV + RV

Vital Capacity (VC) - Max amt that can be forcefully exhaled & max inhalation

Total Lung Capacity (TLC) - Amt of all gas vol in lungs
IRV + TV + FRC
IC + ERV + RV

VC + RV

Stimulus

↑ PCO_2

↓ pH

↓ PO_2

Chemoreceptor

Medullary & Aortic & Carotid bodies

Aortic & Carotid bodies

Carotid bodies

Hypoxic drive: Carotid bodies respond directly to low O_2 dissolved in plasma

Respiration Cont.

Hemoglobin = carries O_2 in the blood
 Each RBC contains millions of hemoglobin
Structure: Each hemoglobin molecule has 4 subunits; each subunit binds to one molecule of O_2

Oxyhemoglobin Saturation $\rightarrow$ measurement of how well lungs have oxygenated blood
 % oxyhemoglobin to total hemoglobin

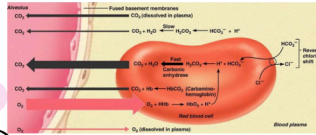
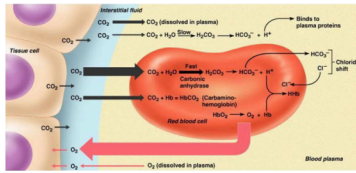
Hemoglobin concentration $\rightarrow$ measures oxygen-carrying capacity of blood

Anemia $\rightarrow$ low hemoglobin concentration;
 = reduced O_2 transport capacity

Polycythemia $\rightarrow$ high hemoglobin concentration;
 can be caused by high altitudes

Erythropoietin = produced by kidneys; stimulates hemoglobin/RBC production in response to low O_2 levels

@ high altitudes
 O_2 supply goes $\downarrow$
 causing more hemoglobin production
 = polycythemia



Carbon Dioxide transport:
 carried in blood by 3 forms $\rightarrow$
 1) Dissolved in plasma (CO_2) 7-10%
 2) Carbaminohemoglobin 20%
 (CO_2 attached to AA in hemoglobin)
 3) Bicarbonate ions (HCO_3^-) Majority

Chloride Shift $\rightarrow$
 Exchange of HCO_3^- and Cl^- across the RBC membrane to maintain ionic balance
 * Bicarb leaves RBC
 * Chloride diffuses in
 Charge remains the same.

@ tissues: $CO_2 + H_2O \xrightarrow{CAD} H_2CO_3$
 $H_2CO_3 \rightarrow H^+ + HCO_3^-$
 Reverse Chloride Shift
 Chloride Shift

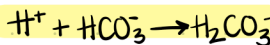
@ lungs: $HCO_3^- + H^+ \rightarrow H_2CO_3$
 $H_2CO_3 \xrightarrow{CAD} H_2O + CO_2$

Reverse Chloride Shift $\rightarrow$
 In pulmonary capillaries
 $\uparrow P_{O_2}$ favors production of oxyhemoglobin

So H^+ dissociates from hemoglobin
 $\&$ recombines $\bar{c}$ bicarb (HCO_3^-)
 to form carbonic acid (H_2CO_3)

Chloride (Cl^-) diffuses out of RBC and bicarb (HCO_3^-) enters

inside the RBC is where H^+ binds $\&$ CAD converts



* Influenced by pH, temp, $\&$ partial pressure of O_2

O_2 loading: O_2 binds to hemoglobin in the lungs

O_2 offloading: Oxyhemoglobin drops off O_2 in the tissues

Dissociation Curve:
 97-75 = 22
 22% O_2 is unloaded to tissues

Veins: P_{O_2} of 40 mmHg
 $\rightarrow$ enough O_2 bind for 75% oxyhemoglobin
Arteries: P_{O_2} of 100 mmHg
 $\rightarrow$ enough O_2 bind for 97% oxyhemoglobin

Curve Shifts = influenced by pH + temp

Bohr Effect: describes effect of pH on affinity of hemoglobin for O_2 $\rightarrow$

higher pH's = hemoglobin binds more readily
 Lower pH's = offloading O_2 ; wants to get rid of it

Alkaline
 More Basic:
 Increased affinity for O_2 $\rightarrow$ better binding in lungs (leftward shift)
 pH $\uparrow$, Affinity $\uparrow$

more Acidic:
 Decreased affinity for O_2 $\rightarrow$ increased offloading to tissue (rightward shift)
 pH $\downarrow$, Affinity $\downarrow$

Lower temp:
 Increased affinity for O_2 $\rightarrow$ better binding in lungs (leftward shift)
 Temp $\downarrow$, Affinity $\uparrow$

higher temp:
 Decreased affinity for O_2 $\rightarrow$ increased offloading to tissue (rightward shift)
 Temp $\uparrow$, Affinity $\downarrow$

Note:

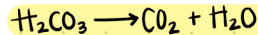
Bonding of H^+ to hemoglobin lowers affinity for O_2

$\rightarrow H^+$ helps blood carry more CO_2

Ex. $\downarrow$ pH = $\uparrow$ H^+ concentration = $\downarrow$ O_2 affinity resulting in inc O_2 offloading

@ the alveoli:

Carbonic Anhydrase (CAH)
 Converts Carbonic Acid back into Carbon dioxide + Water



CO_2 is exhaled (through lungs)

Acid-Base regulation:

Maintained by →

Lungs + Kidneys

quick Δs

Slow Δs

Regulates volatile acids (by breathing)

Regulates NON-Volatile Acids

Ex: Carbonic Acid (H_2CO_3) → because it can be converted into gas & exhaled

Ex: lactic Acid, Ketones, etc → Buffered by bicarb (HCO_3^-) but cannot be regulated by breathing

* Constant range to be maintained: 7.35–7.45

↓
If kidneys detect low HCO_3^- in blood, they will reabsorb vs.
If they detect high HCO_3^- in blood, they will excrete it

Acid-Base Disorders:

Acidosis → pH Below 7.35

Respiratory Acidosis

caused by hypoventilation
Rise of CO_2 (hypercapnia)
 $\uparrow \text{CO}_2 \rightarrow \uparrow \text{H}^+ \rightarrow \uparrow \text{H}_2\text{CO}_3$
Result = pH decreases

$\text{Pco}_2 > 40$

Metabolic Acidosis

caused by excessive Acid production ↑ nonvolatile acids
Loss of bicarb (HCO_3^-) = Diarrhea

$\text{HCO}_3^- < 24$

Alkalosis → pH ABOVE 7.45

Respiratory Alkalosis

caused by hyperventilation
Fall of CO_2 (hypocapnia)
"Blow off" = fast breathing
 $\downarrow \text{CO}_2 \rightarrow \downarrow \text{H}^+ \rightarrow \downarrow \text{H}_2\text{CO}_3$
Result = pH increases

$\text{Pco}_2 < 40$

Metabolic Alkalosis

caused by inadequate Acid production ↓ Nonvolatile acids
Over production of bicarb (HCO_3^-) = loss of digestive acids via vomiting

$\text{HCO}_3^- > 24$

Compensation:

Lungs
Adjust ventilation to alter CO_2 levels

Kidneys adjust bicarb reabsorption or excretion

Normal blood pH =

maintained when ratio of bicarb to CO_2 is 20:1

Henderson-Hasselbalch Equation:

$$\text{pH} = 6.1 + \log \frac{\text{HCO}_3^-}{(0.03)\text{Pco}_2}$$

Effects of Exercise + High Altitude →

Exercise:

↑ O_2 demand + ↑ CO_2 production results in deeper + faster breathing (tachypnea)

Arterial Po_2 + Pco_2 remains stable despite the increased activity

Altitude: ↓ Po_2 at higher altitudes affects oxygen loading →

3 Acclimation Adaptations:

1) Immediate – ↑ ventilation

hyperventilation lowers Pco_2 (respiratory alkalosis)

↑ affinity of hemoglobin for O_2 = Better O_2 loading

Leftward Curve Shift

2) Days – ↑ 2,3 DPG

↑ affinity of hemoglobin for O_2 = Increase unloading to tissues

Rightward Curve Shift

3) weeks – ↑ EPO = ↑ RBCs

= Increased O_2 content in blood

Ventilation can compensate for the metabolic component:

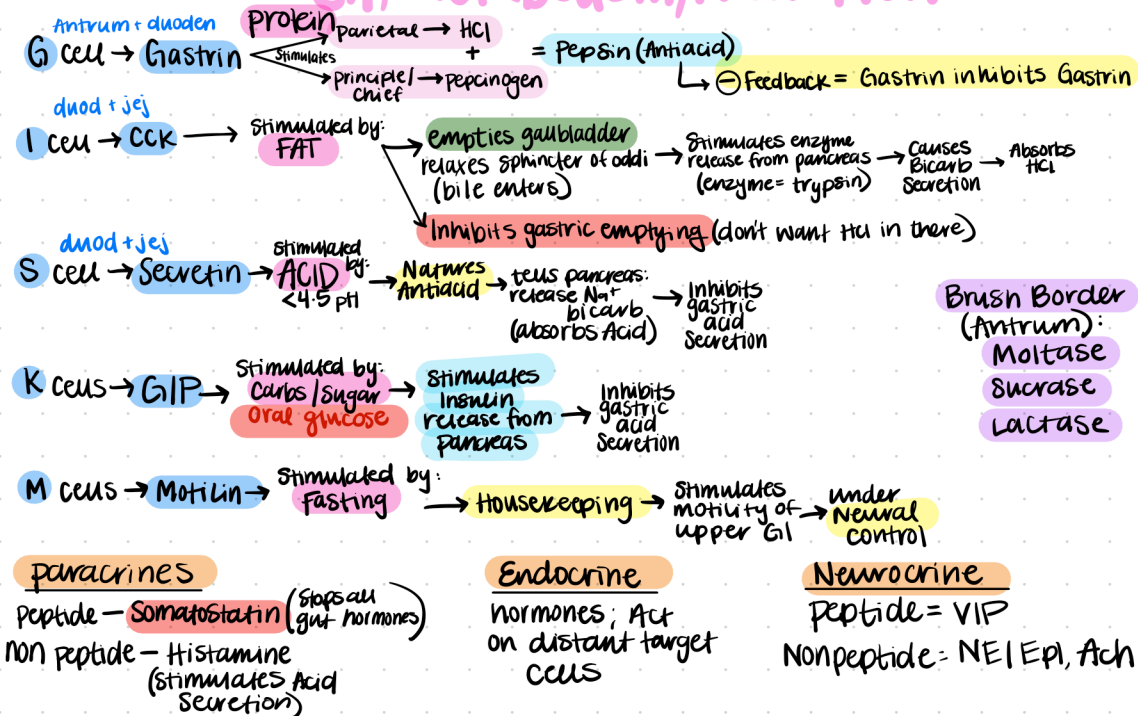
Condition	pH	Pco_2	Ventilation	Cause or Compensation
Normal	7.35–7.45	39–41 mmHg	Normal	Not applicable
Respiratory acidosis	Low	High	Hypoventilation	Cause of the acidosis
Respiratory alkalosis	High	Low	Hyperventilation	Cause of the alkalosis
Metabolic acidosis	Low	Low	Hyperventilation	Compensation for acidosis
Metabolic alkalosis	High	High	Hypoventilation	Compensation for alkalosis

Have to determine if ventilation rates are the cause or the compensation

* Pay attention to Pco_2

Low pH = Always Acidosis

High pH = Always Alkalosis



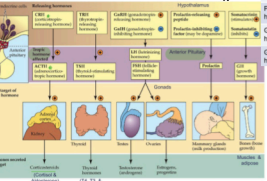
Hypothalamus + Pituitary Gland

Control center:
Secretes releasing & inhibiting hormones

CRH, TRH, GnRH, GnIH, PIH, PRI, GHRH, GHIH

Signals anterior pituitary to release **Tropic hormones** (bring about the Δ that the brain wants)

Have to keep associations straight:



Growth hormone (GH) or Somatotropin

Hypothalamus: ↑ GH/TH, ↑ GHRH
Anterior Pituitary: ↓ GH, ↓ GHIH

Prolactin (PRL)

Hypothalamus: ↑ PIH, ↑ PRH
Anterior Pituitary: ↓ PRL, ↓ PRL

* Works hand-in-hand w/ oxytocin: sucking sensory input (stimulus) causes hypothalamus to produce PRL (milk production) - oxytocin (milk release)
* Dopamine inhibits prolactin - sometimes given for fertility issues/so that women can ovulate normally

ANTERIOR LOBE: ACTH, TSH, FSH, LH, PRL, GH
POSTERIOR LOBE: oxytocin, ADH

FSH/LH: Stimulate testosterone secretion in testes + production of estrogen + progesterone in the ovaries
Hypothalamus: ↑ GnRH, ↓ GnIH
Anterior Pituitary: ↓ FSH/LH, ↓ FSH/LH

once the body has enough androgen, it will tell the hypothalamus to turn off the GnRH

Oxytocin + ADH produced in hypothalamus but released from posterior pituitary

Oxytocin: child birth - stimulates contractions + feedback loop
breast feeding - stimulates milk let down continued by sucking stimulus

ADH: Antidiuretic hormone
Stimulus - ↓ blood volume, ↓ BP, ↑ blood osmolality
Promotes retention of H₂O in kidneys

Central portion: Thalamus good at detecting Hypothalamus Pituitary gland

ADRENAL GLANDS

Outer Adrenal Cortex

Secretes Steroid Hormone in response to ACTH

STRUCTURE:
• Zona Glomerulosa: Aldosterone, Androgens
• Zona Fasciculata: Cortisol, Androgens
• Zona Reticularis: Androgens

Inner Adrenal Medulla: Epinephrine, NE

more so directly innervated by the SNS (fight or flight)

Adrenocortical agents: Note NE gets turned into Epi. which is why they are both present.
* Catecholamines released here = encephalins related to Epi/NE

Hypothalamus secretes CRH → Anterior Pituitary releases ACTH

Adrenal Cortex releases Cortisol, Androgens

Cortisol - the main stress hormone

Functions of Cortisol:
• Blood glucose level
• Immune response
• Stress response
• Blood pressure
• Blood volume

Stress: ↑ ACTH secretion → ↑ glucocorticoid release

The Stress Hormones: Epinephrine (Epi), Norepinephrine (NE), Cortisol

Aldosterone - ALD

Reabsorbs Na⁺ and H₂O in the distal tubule and collecting ducts
Functions:
• Na⁺ reabsorption
• H₂O reabsorption
• K⁺ secretion
• BP
• Blood volume

Stages: 1. Alarm Run, 2. Resistance, 3. Exhaustion

General Adaptation Syndrome: good for proper recovery, Cortisol helps inhibit immune system so it doesn't over respond

Excessive ALD = hypokalemia, Deficient ALD = hyperkalemia

Insufficient glucocorticoid = Addison's Disease

General Adaptation Syndrome: good for proper recovery, Cortisol helps inhibit immune system so it doesn't over respond

THYROID GLAND

* **TSH** stimulates the thyroid. Iodine is taken in through diet & converted to iodine - has to be present for the production of T₄ and T₃

Usually one of the 1st RP jobs are: TSH and Free T₄

TSH will be elevated if: thyroid not working properly, thyroid cancer, thyroiditis

Where is the problem? Too high, Too low, Too high

What would you expect to see with the other components?

Too high: metabolic activity, carb + fat metabolism, protein synthesis, nervous system, HR, ability to sleep, abnormal menses, Graves disease

Too low: metabolic activity, carb + fat metabolism, protein synthesis, nervous system, HR, abnormal menses, Ex. Goiter

Calcitonin: made by C cells of the thyroid - stimulated by TcO₂ - lower serum Ca²⁺ levels

Functions: inhibits osteoclasts (Ca²⁺ release from bone), stimulates excretion of Ca²⁺ in kidneys, inhibits intestinal abs of Ca²⁺

Ex. Goiter: happens when there's no iodine in diet to convert to iodine - so no T₄/T₃ - the TSH secretion is high b/c NO feedback happening

PARATHYROID

Parathyroid hormone (PTH): Secretion increases Ca²⁺ serum levels

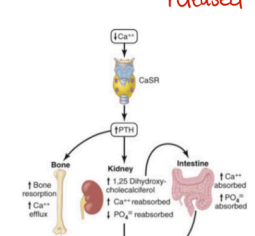
So, if we detect ↓ Ca²⁺, calcitonin released

↓ Ca²⁺ PTH secreted

* Calcitonin is the antagonist to PTH

↑ osteoclast activity, ↑ Ca²⁺ reabsorption in kidneys, ↑ absorption of Ca²⁺ in GI tract

Also PTH ↑ production of Vitamin D₃



Feedback Loop

PANCREAS

Endocrine + Exocrine gland

cells located in the pancreatic islets.

α cells = glucagon
β cells = insulin

helps mobilize glucose in bloodstream to help ↑ blood sugar

primary hormone regulating plasma glucose concentration

Detected in the Liver

Insulin helps glucose get into tissue cells

Insulin is anabolic hormone

→ causes tissue growth

- Once insulin binds @ cell surface — glucose transporters translocate to cell surface — which allows glucose to come in

* Important because normally cells are impermeable to glucose

helps to lower blood sugar into normal range

- ① binds to receptors on target cells
- ② indirectly stimulates enzyme glycogen synthase to promote sugar storage (muscle + liver)
- ③ stimulates fat storage (adipose tissue) for long-term storage

Glucagon: Antagonist to insulin

Secreted when blood glucose levels are low

Detected in the Liver, Secreted from Pancreas

- ① Stimulates liver to hydrolyze glycogen into glucose
- ② releases into blood
- ③ Gluconeogenesis stimulated: conversion of noncarbs → glucose
- ④ Stimulates lipolysis in adipose tissue → fat released & used as fuel source (instead of glucose)

Important tissues:

Muscle
Liver
Adipose tissue

Diabetes -

Type 1: Lack of insulin

Type 2: insulin resistance (due to ↑ blood sugar for so long)

ANABOLIC

↑ Blood Sugar → α insulin → ↓ Blood Glucose to storage

CATABOLIC

↓ Blood Sugar → β glucagon → ↑ Blood Sugar from liver storage

Hormones help regulate body metabolism growth + reproduction

Antagonistic — hormones working in opposite direction
Ex: insulin / glucagon

Synergistic — work together to produce effect
Ex: prolactin / oxytocin

Permissive — one hormone making target cell more responsive to 2nd hormone
Ex: Estrogen + progesterone

Hormones have 1/2 life time to lose 50% of something. Usually hormones are removed from the blood by the liver

higher plasma concentrations of hormones causes effects different from normal

Body Response — prolonged exposure → decreased # of hormone receptors = **Downregulation**

Fewer receptors = fewer hormones can bind = less effect / response

Proteolytic mechanism

up regulation / priming effect if we give drug that ↓ activity of receptors the body will increase the # of receptors to become more sensitive & be more likely to bind to target hormones

KIDNEY

Secretes **Erythropoietin (EPO)**

Stimulates production of RBC by red bone marrow (flat bones — sternum, hip, vertebrae)

PINEAL

Secretes **Melatonin**

Regulated by the hypothalamus

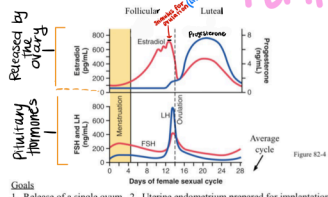
- stimulates production when it gets dark → regulation of circadian rhythms
- production inhibited during day time

THYMUS

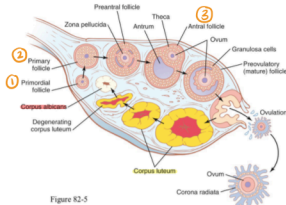
Secretes **Thymopoietin**

increases WBC production in lymph nodes

REPRODUCTION FEMALE REPRODUCTION



- Goals
1. Release of a single ovum
 2. Uterine endometrium prepared for implantation



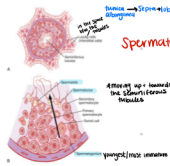
*side note: Corpus luteum degenerates & becomes **Corpus albicans**
Scar tissue

1. **Premordial** - Single layer of granulosa cells
2. **Primary** - Multiple layers of granulosa cells
Ovum enlargement, Regulated by LH, FSH
3. **Antral** - Fast granulosa cell proliferation
→ producing follicular fluid
→ Spindle cells develop into **theca**

Theca Interna: Secrete sex hormones
Estrogen, Progesterone

Theca Externa: Vascularized capsule

male reproduction



Spermatogenesis = Complete process in males
Result of meiosis is 4 haploid cells
germ cell
1/2 # of chromosomes will be: X or Y

Composition of Ejaculate:

- Semen = **Bucc** pH → Sperm don't live
- Acidophilic acidic environments

- Capable of maturing within 18-24 hr of storage in the epididymis

- **Seminal vesicles** - mucous containing Fructose, Citric acid, Prostaglandins, Fibrinogen

- **Prostate gland** - milky fluid containing Ca²⁺, citrate, non prostatic acid, prostatic acid

- **Semen** - Fluid + Sperm pH: 7.5 (vaginal pH is 4)

- Sperm:**
- Mature after ejaculation
 - 400 million Sperm / ejaculate
 - Storage up to 1 month in the epididymis
 - Likes: Neutral to basic pH & 2°C under body temp

Testosterone important for:

- Enlargement of primary sex characteristics
- Hair growth, deep voice, ↑ skin thickness
- ↑ bone density, ↑ bone mass, ↑ Baldness
- production starts @ about 10 years

Key:

- Corpus Luteum = Yellow
- Corpus Albicans = Red
- FSH = causes follicle to mature
- LH = causes egg to be expelled (ovulation)

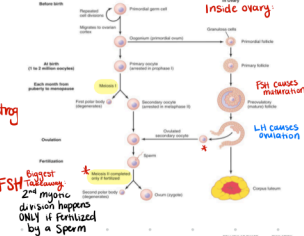
1. FSH secretion causes follicle to mature
2. As follicle matures it secretes **Estrogen**
3. Estrogen increasing triggers quick increase of LH + FSH (but mostly LH)
4. LH peak surge induces ovulation = ovum expelled from follicle
5. LH surge also stimulates production of the corpus luteum
6. Corpus luteum is the only thing that produces **progesterone**
→ Also produces some levels of **estrog**
7. Progesterone prepares uterus endometrium for implantation
8. Estrogen imposes ⊖ feedback on FSH
→ released into bloodstream & tells FSH to slow down

If there is no implantation -

9. Corpus luteum degenerates into corpus albicans b/c there is NO FSH to maintain it
10. No more corpus luteum = NO more **Progesterone** + **Estrogen**
the decline allows FSH secretion to start back up

11. Menstruation
12. Repeat

OOGENESIS



Males:

Spermatogenesis prokaryotic only after fertilization

Meiotic divisions of a primary Spermatocyte produce 4 mature Sperm

Products of meiosis (sperm) undergo additional division in the maturing process

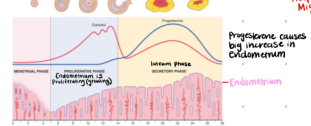
FSH causes maturation

LH causes ovulation

Progesterone causes progesterone stimulation (hormonal) (increases & deviation) in the uterus. Controlled by FSH. (more similar to males. L stimulates FSH) (FSH stimulates Progesterone)

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Estrogen Effects:

- ↑ cell proliferation
- tells vaginal epithelium to become stratified (from cuboidal state)
- ↑ Uterine size
- ↑ thickened epithelium of fallopian tubes
- ↑ Stroma breast tissue (lead to tenderness)
- ↑ Osteoclastic activity
- Tells cervix to secrete watery slim (Sperm can swim)

Progesterone Effects:

- ↑ Secretory & in endometrium
- ↑ Uterine contraction (very small amount in there)
- ↑ Secretion in fallopian tubes
- ↑ Breast glands & milk
- Tells cervix to secrete a thick viscous secretion
- ↑ body temp by 0.5°C

Puberty:

gradual ↑ GnRH in 8th year
Onset b/w 11-16 years old

Menarche 1st cycle
Avg onset <13 years old

Menopause 40-50 years old
Irregular cycles
Low Estrogen/Progesterone

Regulation of Spermatogenesis:

LH: Stimulates Leydig cells to secrete testosterone

Testosterone: ↑ growth/division of germinal cells

FSH: Stimulates Sertoli cells (support for spermatogenesis)

Estrogens: made from testosterone & FSH stimulates Sertoli

Gn: Promotes division of spermatogenesis

→ Needed for sperm production

Inhibin: Produced by Sertoli cells
○ Feedback regulation of FSH

Male Sex Act:

Stimulus - glans penis has sensory nerves + psychic stimulation

Parasympathetic - POINT vasodilation

↑ blood flow to corpus cavernosum & spongiosum

Sympathetic - STROT contraction of vas deferens Ampulla, Prostate, & seminal vesicles

Major Sex Hormone: Testosterone

is a bound Steroid hormone/derived on cholesterol

So it's NOT Active circulates bound to plasma protein (albumin is the most common)

convert to: **Dihydrotestosterone** (free hormone) = Active

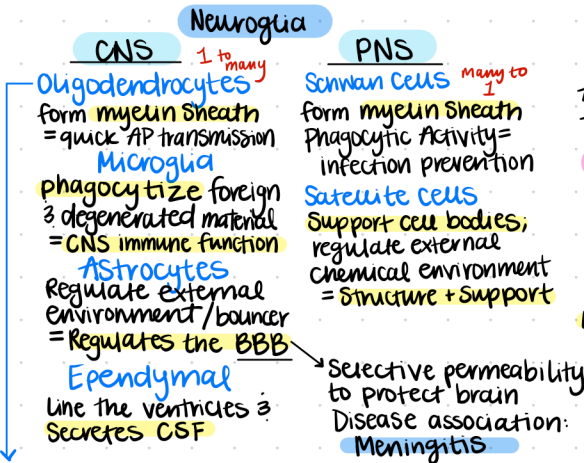
metabolized by: **The Liver**

Sensory
Afferent signals → Spinal cord → Brainstem → Cerebellum → Thalamus → Cerebral Cortex

Motor
Efferent signals → CNS → Muscles/Glands

Levels of control:
Lower regions = automatic
Higher regions = deliberate
Major levels of CNS function:
Spinal cord level = signal transmission / reflex arcs
Lower brain level = (subcortical / lizard brain) subconscious activities i.e. breathing, emotional patterns
Higher brain level = (cortical) memory storehouse, conscious control, critical thinking

Most Memory Storage happens here
↓
Repetitive synapse firing solidifies memories



APs = Nerve Impulses
Types of Synapses →

Electrical:
Bidirectional transmission via gap junctions (NOT as common)

Chemical:
Unidirectional transmission
Neurotransmitters that excite or inhibit

Synaptic vesicles contain the neurotransmitters
Voltage-gated Ca^{2+} channels
More Ca^{2+} = more vesicle release

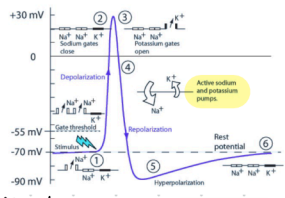
Synapse = Allows mvmt of ions

Receptor proteins:
Gated ion channels
Cation - Excitatory Channels Na^+, K^+, Ca^{2+}
Anion - Inhibitory Channels mainly Cl^-

Others:
Glycine - inhibitory in Spinal cord
Histamine - H1 receptor Effects arousal (Ex. Benedryl = histamine inhibitor, makes you sleepy)

Nitric oxide - behavior & long term memory

Cannabinoids - CB1, CB2 receptors, mood, appetite, nausea, vomiting, cognition, memory



Na^+/K^+ pump = 3 Na^+ Out, 2 K^+ in
Kicks in during Repolarization
RMP = -70; More Na^+ Outside, More K^+ Inside

ARP = During AP
 Na^+ channels inactive (Not just closed)
RRP = K^+ channels still open
Strong stimulus can overcome this (i.e. Seizures)

Synaptic excitation:
Summation
Spatial → Multiple neurons Sum up to 1 AP
Temporal → Many signals from ONE neuron until threshold met
Inhibitory neurons @ the axon have greater effects than excitatory neurons further away

Myelin sheath gives Axons their white color = White Matter
Grey matter = cell bodies + dendrites = NO myelin sheath
Disease Association: Multiple Sclerosis = Axons Striped of myelin sheath

	Synthesis	Metabolism	Uses	Receptors
ACh	Acetyl CoA + Choline (Enzyme: Acetylcholinesterase)	Acetylcholinesterase (Enzyme recycled by Presynaptic Neuron)	Somatic + Autonomic (SNS: PNS)	Cholinergic: Nicotinic & Muscarinic
NE	Made from tyrosine (in diet: cheese, meat)	COMT, MAO	SNS (usually excitatory)	Adrenergic (α, β, P, B_2)
Dopamine	Made from tyrosine (in diet: cheese, meat)	COMT, MAO	Substantia Nigra secretes Nigrostriatal - MMT, Mesocortical - Reward, Tuberofundibular - Outputs to pituitary	D_1 - stimulates, D_2 - inhibits
Serotonin 5-HT	Made from tryptophan	MAO	Alters mood/role in Anxiety, Nausea, Vomiting, Migraines	Many
GABA	Made from Glutamate (+ B_6)	GABA transaminase	Major INHIBITORY neurotransmitter Hyperpolarizes Post-synaptic neurons	GABA-A (opens Cl^- channels) = cell more $-$ = harder to have AP
Glutamate	Made from Glutamine	GABA	Major EXCITATORY neurotransmitter Excitotoxicity can occur & overstim Alzheimer's, Cell suicide	NMDA, AMPA Metabotropic receptors

protective mechanism:
Fatigue - prevents neuronal damage
Important for Seizure termination

Effects of pH: ex. Hyperventilation (panic attacks) → F/F response → CO_2 ↓ → Blood pH ↑
Alkalosis ↑ excitability
Acidosis ↓ excitability
pH > 7.4 = Alkalotic

Neuro Intro Cont.

Oxygen = high requirement
for neuronal excitability
Hypoxia / No cerebral bloodflow
for 3-7s leads to unconsciousness

Receptor potentials: Frequency of
repetitive APs determine sensation

→ one signal may not be enough,
but if many signals coming
in → brain gives it priority

Phasic receptors =

good for sensing Δ ; fast adapting
ex. feeling clothes you wear

Tonic receptors =

Slow adapting; maintain high
firing rate as long as stimulus
is applied

ex. hand on stove - continuous signal
until stimulus removed

baro/chemo receptors in CV system:
respond to Δ in pTT / Δ in CO_2

Degree of Accommodation is related
to the type of tissue stimulated

Fast pain

< 0.1s $\bar{P}$ stimulus

Sharp, Acute

Skin only



Neospinothalamic tract
tactile receptors in dorsal
column tract = hand on stove
• Release glutamate
(Excitatory!)

Analgesia System:

Natural Anti-pain
pathway

• Serotonin → Stimulates
Enkephalin release
||
Natural Opioid

Slow pain

> 1s $\bar{P}$ stimulus

Aching, chronic

Skin + deep tissue

★ Substance P involved

Nociceptors adapt
very little &
can become more
sensitive overtime
= Hyperalgesia
(more easily triggered)



Paleospinothalamic
tract:

Poor ability to locate
• Release glutamate +
Substance P
helps sensitize nerves
to feel pain more acutely

Sensory categories:

Chemoreceptors - Chemicals (also in taste & smell)

Photoreceptors - Light

Thermoreceptors - Temperature

Mechanoreceptors - Mechanical deformation (hearing, touch)

Nociceptors - Pain (temp, pressure, chemicals)

Proprioceptors - Mvmt/balance (muscle, tendons, joints)

Special senses - Equilibrium (vision, hearing, taste, smell)

Cutaneous receptors - Skin

Merkel's discs: Indentation

Root hair plexus: Light brushing

Pacinian Corpuscle: Vibration; deep pressure

Meissner's corpuscle: Change in texture

Free nerve ending: touch, pain, hot, cold

Ruffini ending: Skin stretching

Sensory Pathways:

Dorsal Column-Medial
lemniscal system:

Fast transmission

High Spatial Orientation
touch detection & high
locality & fine gradation
of intensity

Crossover in Medulla

Think: fingertips

Anterolateral System:

Slower transmission

Detects broad sensory

Spectrum-

Pain, temp, crude touch

Crossover in Spinal column

Think: tickle/itch

Somatosensory areas →

Areas & larger areas in somatic
cortex have larger # of sensory
receptors

Greater density
of nerves in sensitive
areas

= face + hands

Lower 2pt touch
threshold

Less sensitivity in
larger areas

= back + abdomen

higher 2pt touch
threshold

Recruitment process for signal intensity

Lateral inhibition for fine differentiation

Thermal sensations:

Cold receptors more numerous

Nociceptors detect extreme hot or cold

Temp Δ respond more than static temp

19, 20

Vision/Special senses

7/.

Visual pathway:

Cornea → Anterior chamber → pupil → (aqueous humor) posterior chamber → lens → (vitreous chamber) posterior cavity → Retina

Iris = outer radial muscle + inner circular muscle
Can increase/decrease diameter of the pupil (CN III)

Constriction
Parasympathetic stimulation
Circular muscle contract

Bright light response

Dilation
Sympathetic stimulation
Radial muscle contract
MYDRIASIS

Dim light response

fovea = most cones
optic disc = blind spot
peripheral vision = more rods

Retina = where the photoreceptors live

Located in the pigmented choroid (dark/contains melanin) - good b/c this means it helps to absorb lots of light

Lens = Avascular
sits behind the iris

lens is attached to suspensory ligaments (zonules)

which are controlled by: ciliary muscles/bodies

These produce: **Aqueous humor**

which fills the Ant/post chambers of the Ant cavity provides nourishment to lens + cornea

Drains into the Canal of Schlemm & then back into blood

The lens helps to bend the light coming in:

Light Refraction - image is flipped upside down & Left to Right

Lens Accommodation:
keeps objects focused on the retina as the distance b/w eye & object moves

FAR VISION

Ciliary muscle relax
Suspensory ligaments tighten
lens thins & flattens

Sympathetic**NEAR VISION**

Ciliary muscle contract
Suspensory ligaments relax
lens thickens & rounds

Parasympathetic**Myopia** = Nearsightedness

Elongated eyeball

Distant images brought to focus in front of the retina

Correction: **Concave lens****Hyperopia** = Far sightedness

Short eyeball

Distant images brought to focus behind the retina

Correction: **Convex lens**

Snellen eye chart measures visual acuity: near point / sharpness of vision / resolving power

Astigmatism = irregular curvature

Several points of focus on the retina

Correction: **cylindrical lens****Presbyopia** = age-related

reduced lens flexibility

Cataracts = cloudy lensCorrection: **surgical removal****Visual pathway:**

Retina → pigmented cells
photoreceptors

Horizontal cells - send signals laterally for an inhibitory effect: when focusing on one spot, will inhibit other points

bipolar cells

Amacrine cells

Ganglionic cells

Optic nerve
CN II

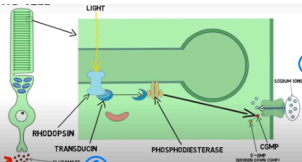
Rods: B & W vision in low light

contain **Rhodopsin** (purple pigment)

which absorbs green light best

* Requires Vitamin A *

Phototransduction: occurs in Retina - light is converted into electrical signals



① Ion channels activated by cGMP - allow Na⁺ to flow into rod cells

② Na⁺ inflow causes continuous depolarized state = continuous release of glutamate (excitatory ntm)

③ Rhodopsin is made up of retinal - which is what absorbs light (Vit A necessary)

④ when light is absorbed, protein transducin is activated

⑤ which then activates phosphodiesterase

Cones: Blue, Green, Red

Color blindness - X-linked

Loss of **Red** cones: **Protanopia**

Loss of **blue** cones: **tritanopia**

Loss of **green** cones: **Deutanopia**

⑦ Ion channels begin to close

⑧ Cell becomes hyperpolarized (K⁺ leaving, No Na⁺ entering)

⑨ glutamate release decreases

⑩ Acts as a signal that a light stimulus is present

19,20 Vision / Senses cont.

more on color blindness:
failure of 2 types of cones
= **Monochromacy**
To detect color blindness -
ishihara test

Eye movement:
LRG SO4(3)

Fixation movements:

Voluntary = fixate on chosen point
frontal cortex control

Involuntary = maintain fixation
occipital cortex control

Saccadic Movement = quick, random
movements to keep new
visual inputs

Converts sound vibs
into electrical inputs

Stapedius muscle
dampens Stapes
if sound is too loud

Inner ear = the cochlea

Scala Vestibuli
Portion of bony labyrinth

Scala Media (ENDOLymph)
Cochlear duct

Scala tympani

Hearing Impairment:

Conductive: Wax, fluid,
Otosclerosis (bone overgrowth)

Sensorineural: hair cell damage

Presbycusis: age-related

Smell:

CNI olfactory nerve =
Short transmission to
brain

Olfactory receptors - bipolar
neurons ciliated dendrites
projecting into nasal cavity.

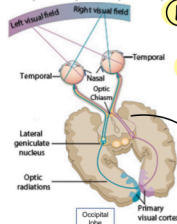
Proteins bind to odors
1 odorant binds to 1 protein
(specific receptors)

Olfactory nerves = **unmyelinated**

Olfaction greatly influences taste

Interactions c the
thalamus & limbic system

Visual pathway:



① visual field is projected onto
the **R** side of the retina in
both eyes - then gets transferred
to the **R** lobe of the occipital cortex

Medial aspects crossover at
the **optic chiasm**

Optic nerve → optic chiasm

Lateral geniculate nucleus
(**THALAMUS**) ← Optic tract

→ **Primary Visual Cortex**
(occipital lobe of cerebrum)
processes initial
visual information

↓
Secondary Visual Cortex
integrates motion
3D position, color,
and details

Hearing:

Middle ear =
blw TM and cochlea

contains 3 bones
(ossicles)

• **Malleus** → vibrations
transmitted along the bones
• **Incus**
• **Stapes** → attached to oval window

High frequency = terminate early
Low frequency = travel further

Sound waves enter
Scala media

tectorial membrane
vibrates

Stereocilia
bends

K⁺ returns
to perilymph
@ base of
the stereocilia

K⁺ moves
in =
Depolarization

K⁺ channels
facing the
endolymph
OPEN

Contains
organ of
Corti = converts sound
waves into
electrical impulses
= **vestibulocochlear nerve**
(**CN VIII**)
Contains
hair cells - located on
basilar membrane, projecting
into endolymph of cochlear duct

Taste: gustation
receptors = taste buds
taste pathway:
CN VII + CN IX

Medulla oblongata → Thalamus

gustatory cortex

taste Receptors:

Salty = Na⁺ depolarizes
Sour = H⁺ depolarizes

Needs Ca²⁺
to release
ntm

Sweet & Meat =
sugar/glutamate
binds & activates
G-proteins → 2nd messenger
close K⁺ channels

Bitter = quinine binds
& activates
G-protein → 2nd messenger
releases Ca²⁺
into cell

Swallowing
Coordinated by:
CN V VII IX X XI XII
5 7 9 10 11 12

types of papillae:

Fungiform (Salty + Sweet)
Anterior surface

Circumvallate (Bitter)
Posterior surface

Foliate/Filiform (Sour)
Sides

CN VII
Facial

CN IX
glossopharyngeal

CN IX
glossopharyngeal

Salivary glands:

Sublingual
CN VII
Facial

Submandibular
CN VII
Facial

Parotid
CN IX
glossopharyngeal

Interneurons = found in cord grey matter

Integrate spinal cord function

Sensory signals/signals from brain
more likely to terminate on interneurons

Responsible for sending motor message out

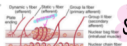
Renshaw cells → focusing of signals **LATERAL INHIBITION**, fine motor control ↑
Propriospinal fibers → coordinate mvmt in various areas/limbs

Muscle Spindles: Stretch detectors **Golgi tendon organs** = tension detectors

Central region has NO Actin or Myosin

→ **Unable to contract by itself**

Sensory primary: encircles center of fiber
Secondary: 1-2 smaller fibers branching out



α = Stimulate muscle contraction @ motor end plate
Extrafusal fibers

γ = control muscle spindle tone; regulation
Intrafusal fibers

Voluntary Activity:

α-γ co-activation =
Initia = extrafusal fibers contract
@ same time

Prevents receptor (sensory) from
activating & causing reflex arc
Maintaining dampening fcfn

AS extrafusal fibers are
pulled apart, intrafusal
fibers are stimulated
& they oppose the reflex

Central control of γ motor neurons helps
to stabilize joints → good for anti-gravity
muscles & muscle movements under load

↑ tension = relaxation of corresponding muscle
(stretching)

Stretch → firing
Shortening → firing

These occur when spindle
length is 1.5x rest length
→ static & dynamic firing

Dynamic: Sudden AS
Primary fibers stimulated
Static: Slow & Steady AS
Primary & Secondary fibers stim.

Reflex Arc:

quick heavy book dropped into arms
MUSCLE SPINDLE STRETCHED → Sensory firing ↑

causes contraction of same
muscle (α)

Spindle no longer stretched

Afferent to spinal cord

Inhibitory interneuron branches out

Sends inhibitory motor signal to (inhibits)
antagonist muscle

causing relaxation

Flexor reflex = Nociceptor reflex
(elicited by pain)

*quick! happens within spinal cord

Primary Motor Cortex:

Stim of neurons excites patterns of separate muscles

Premotor Area:

Stim activates complex patterns of mvmt

Develops "motor image" → sends to PMC
or basal ganglia to be carried out

Supplementary Motor area: (side 20)

Stim causes bilateral mvmts

Winks & premotor area for "positioning"
head rotation, eye mvmt, etc

Sensory fibers feed
into Motor Cortex to
tell it what they want to
happen.

Transmission of signals to muscle
crosses spinal

or pyramidal tract
→ terminate on interneurons

Accessory pathways:
Basal ganglia

more complex mvmt
Cerebellum

Red Nucleus:

Alternative pathway - Cortical signals → spinal cord

Skips cerebellum & basal ganglia

Used for discrete activation

of tiny muscles (fingers/hands)
(small in humans bc we don't use it)

Brainstem:

Medulla, pons, mesencephalon

control involuntary functions

helps control signals that make
it back down to the muscles

Pontine reticular nuclei = Excitation
Medullary reticular nuclei = Relaxation
(ANTAGONISTIC)

of anti-gravity muscles
Live going from sitting → standing

Vestibular nuclei: Detects stimuli & reports to Pontine reticular nuclei to maintain equilibrium

Someone throws your wheel → Vestibular stimulated → Pontine reticular nuclei activated → Anti-gravity muscles stimulated → helps you stay standing

Cerebellum → "Silent area"

- Useful for smooth progression
- Makes corrective adjustments
- Plans sequential mvmts
- learns mistakes = muscle memory

- ★ Vermis - Axial body
- ★ Intermediate - distal upper & lower limbs
- ★ Lateral zone - joins & cerebellar cortex

Assists in planning sequential mvmts (i.e. dancing)

Neuronal Circuits of cerebellum:

- Where are signals coming from? **INPUT pathways**
- Corticopontocerebellar pathway** - from areas of brain making the decisions
- Olivocerebellar tract** - from motor cortex, basal gang, spinal cord
- Vestibulocerebellar fibers** (detecting mvmt/equilibrium) - vestibular apparatus
- Reticulocerebellar fibers** - brainstem reticular formation

Sensory
Afferent pathways from periphery -

DORSAL Spino-cerebellar tract - tells cerebellum what's going on to body
muscle contraction, tendon tension, positions, forces

VENTRAL Spino-cerebellar tract - tells cerebellum what signals have been received by Ant. horns (which go on to activate mvmt)

Every signal entering
Cerebellum splits & goes
↑ to nucleus & cerebellar cortex

Deep cerebellar nuclei:

Dentate: Coordinate sequential mvmts

Cerebrocerebellum

Interposed: Coordinates antagonistic muscle mvmt

Spino-cerebellum

Fastigial: Equilibrium - AS steps are carried out: makes sure you don't fall

Vestibulocerebellum

Cerebellum
strengthens
signals
from cortex
ex. coordinates
on/off for
antag muscles

*** Cerebellar Cortex**:

• gets a lot of excitatory
afferent inputs coming in

• **Purkinje cells are inhibitory**

* which prevents
overshooting

* they learn to correct
motor errors *